Luiss
Libera Università Internazionale
degli Studi Sociali Guido Carli

Lab on Docker

Data Privacy and Security

Data Science and Management (DASMA)

LUISS

Docker baseline

Docker is a container platform that allows developers to package applications and their dependencies into lightweight, portable environments that can run consistently across different systems.

Main Docker concepts

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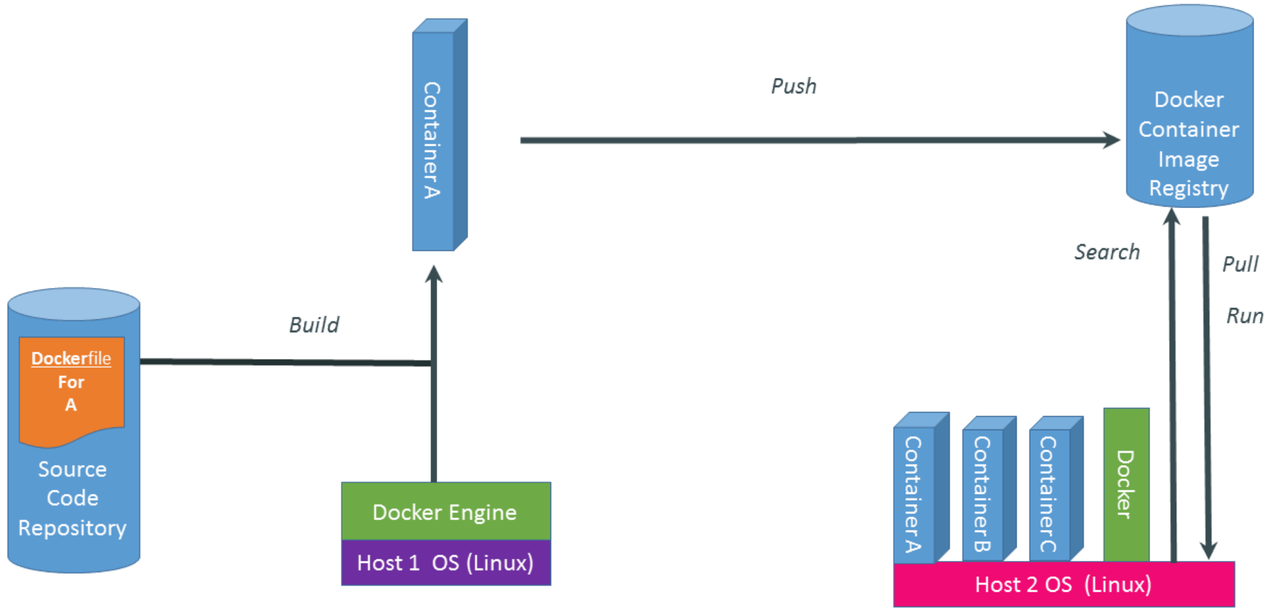
- **Docker Image**: A read-only template with instructions for creating a container (like a recipe)
- **Docker Container**: A running instance of an image (like the dish)
- **Dockerfile**: A text file with instructions to build an image
- **Docker Hub**: A public registry for sharing images

How Docker Works



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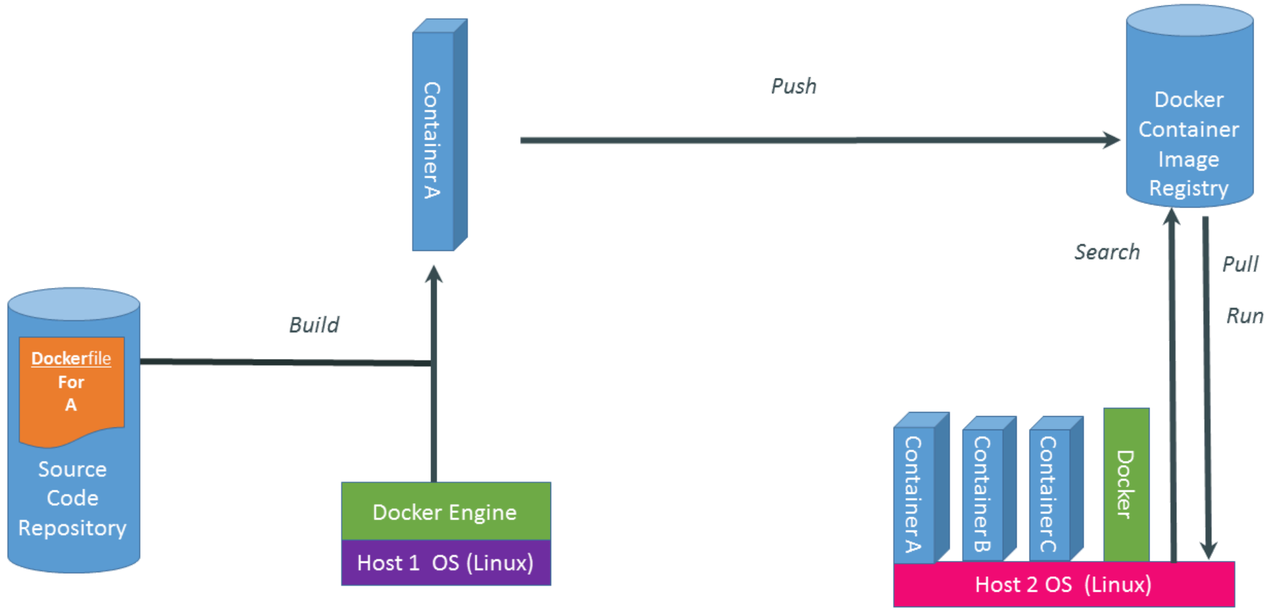

1. A **Dockerfile** describes how to build an image
2. `docker build` creates the image from the Dockerfile
3. `docker push` uploads the image to a **registry** (e.g., Docker Hub)
4. On another machine: `docker pull` downloads the image
5. `docker run` starts a container from the image

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`docker cp` copies files between the host and a container:

```
# Copy a file FROM the container TO the host
docker cp my-web:/etc/nginx/nginx.conf ./nginx.conf

# Copy a file FROM the host TO the container
docker cp ./index.html my-web:/usr/share/nginx/html/
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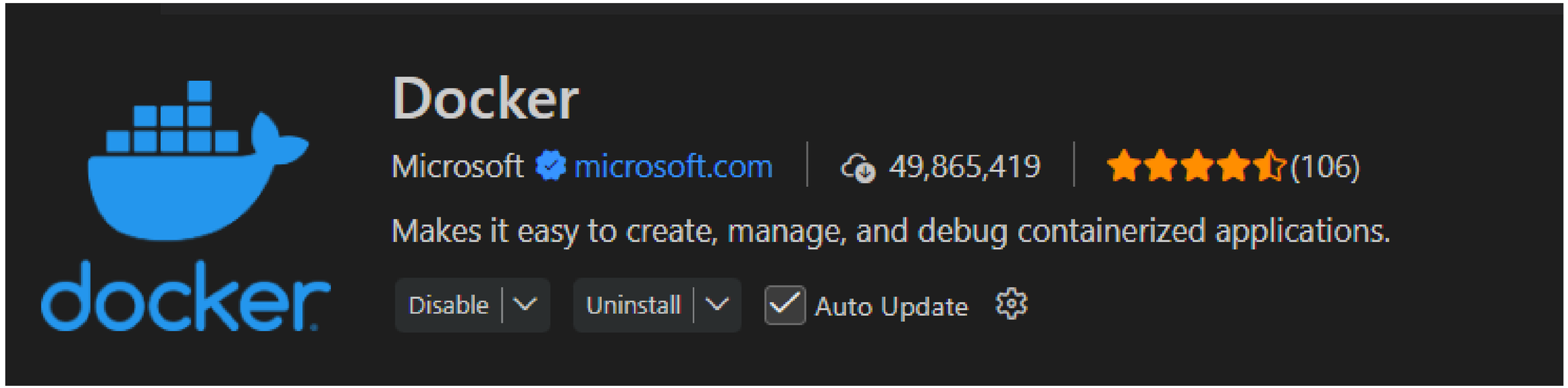
This is useful for:

- Extracting logs, data, or configuration files from a container
- Injecting configuration or data files into a running container
- Debugging: inspecting files inside a container without exec

Docker and VSCode

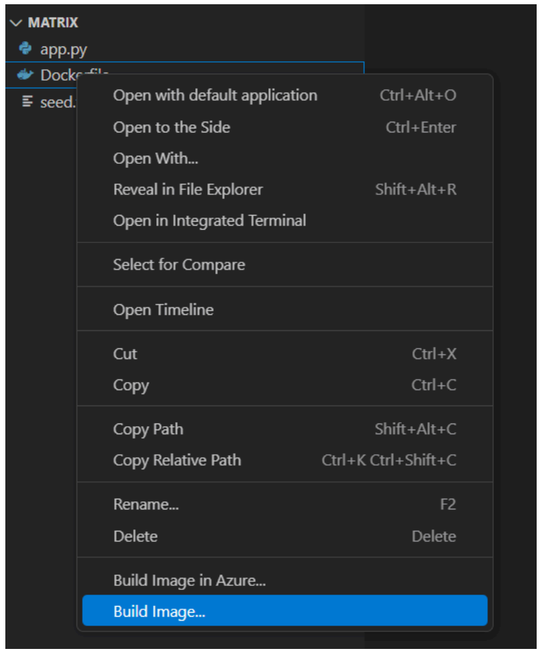
The official Docker extension for VS Code makes very easy to handle containers



Docker and VSCode

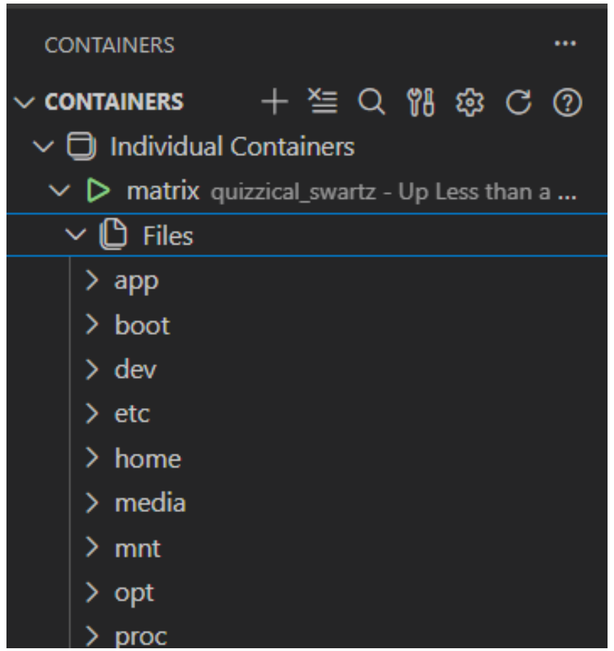
Docker and VSCode

Building and image could be simple as clicking a button



Docker and VSCode

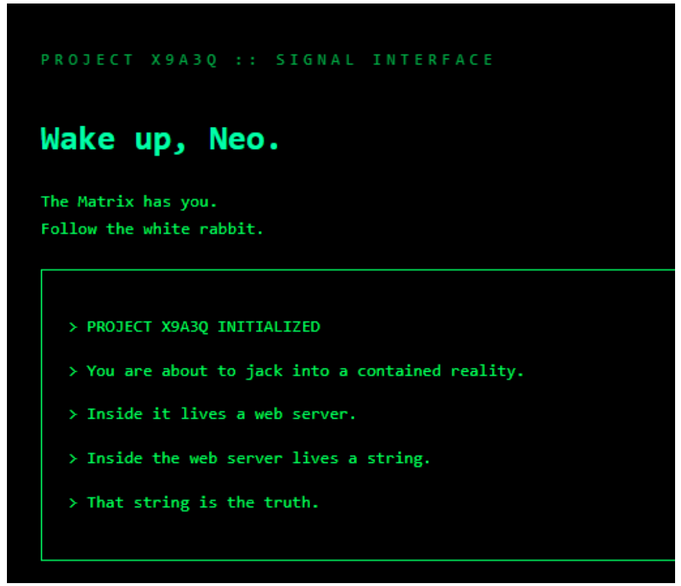
Browsing the container filesystem is also very user friendly.



Project X9A3Q

Docker Challenge 01

Project X9A3Q - Reconnaissance



What can we observe? The challenge...

- let you download a ZIP file;
- require you to build a docker container and take a message;

Project X9A3Q - Hypothesis

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> The fastest way is probably a combination of both

Project X9A3Q - Exploitation

1. **Download the challenge file** from the website.

2. **Read the files** to understand the URL to generate the message.

3. **Run the container** and generate the message.

4. **Find the message** reading the source code and understanding where it is saved.

Project Jarvis

Docker Challenge 02

Project Jarvis - Reconnaissance

What can we observe?

- A system to download the runtime of a local AI.
- You need to obtain the "master" key of the system.

Project Jarvis - Hypothesis

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> Just run the pipeline